

True Effort Score

A transparent, multi-signal model of running load on Apple Watch

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SUMMARY

True Effort Score (TES) measures athletic effort from six signals from Apple Watch. These include heart rate, pace, cadence, grade, vertical oscillation and altitude. A baseline model learns the heart rate of a runner's anticipated workload and reports the gap. A classifier model also labels each vector based on how the runner is moving. As a result, efforts that require significant biomechanical effort (e.g., descending downhill) also count as hard work. Designed for watchOS, TES is built in Swift as a model within the open-source Quiver framework.

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1 History

In the 1970s researchers at Simon Fraser University in Burnaby, British Columbia, Canada, proposed a model of how training performance calculations should work. The idea was based on **training** being the input and **performance** the output. Each workout is reported as an “impulse” that produces two responses: fitness, which builds slowly and lasts, and fatigue, which builds quickly but eventually fades. Performance on any day is roughly fitness minus fatigue. This became known as the fitness-fatigue model, or the impulse-response model. To manage it they defined the term **training impulse** or TRIMP to determine the significance of each workout. The first version scored a swimmer’s training in points for every 100 meters, weighted by intensity. By 1990, the number came from heart rate: the minutes of a workout, weighted by cardiovascular intensity.

Over the years the concept was refined. In 1993, a simpler scheme called **summated heart-rate zones** appeared in coaching literature. The possible range of heart rate was divided into five zones, and the minutes spent in each zone were multiplied by the zone’s number from one to five.

Introduced in 2001, a method named **session-RPE** determined training load without relying on heart rate data. Building on a ten-point perceived exertion scale from the 1960s, the calculation involved multiplying the athlete’s self-reported exertion by the session’s duration in minutes. What this approach sacrifices in biometric precision, it gains in universal reach. It applies to any sport, requires no hardware sensors, and captures systemic fatigue that heart rate alone cannot measure.

1.1 The running boom

Fast forward to today and most fitness watches calculate training load from one or a combination of these ideas. Heart-rate zones remain the default on fitness watches, while perceived effort can be seen in the post-workout rating many apps now request. In many ways both approaches utilize the TRIMP’s formula of intensity multiplied by time.

In many respects, TRIMP provides a good approach to measuring cardiovascular effort. However, the sport of running has grown in both depth and participation since the 1970s. Today, even though less than 1% of people have ever run a marathon, 2027 will mark the first occasion where a world marathon major (London) spans two days, welcoming 100,000 participants across the weekend. For the first time in history, we are seeing runners complete the 26.2-mile distance in under 2 hours. And in the trail running world, long-standing records at the Ultra-Trail du Mont Blanc continue to fall, with runners now completing the roughly 108-mile course with over 32,000 feet of elevation gain and loss in under 19 hours.

2 Understanding true effort

While advancements in both nutrition and shoe technology have certainly contributed to running's success, arguably the biggest advancement has come from training principles. As runners know, successful training requires balancing effort and recovery. Knowing how hard to push on any given day dictates the difference between success and injury. Today, champion runners develop not only cardiovascular excellence but also muscular endurance, extending to their tendons and ligaments. Examples include:

- **Running downhill.** On a steep descent the quadriceps contract eccentrically to brake every foot strike. Compared to running on a flat surface, a -9° (about 16%) slope produces roughly 50% more impact force and over 70% more braking force with each stride [9], and the muscles, ligaments and tendons absorb that added load. Because gravity supplies much of the work, the effort may not register as significant cardiovascular activity [10].
- **Interval training.** Short periods of intense effort are separated by walking, standing, or easy recovery jogging. While surges demand significant cardiovascular output, averaging them together with the rest periods masks the true intensity.
- **Technical terrain.** Rocks, roots and loose footing demand concentration, agility and flexibility on every stride, whether climbing, descending or on the flat. Pace slows and heart rate may stay moderate, so neither reflects the added work.
- **Power-hike.** On a steep climb a runner may drop to a hike. Pace and cadence collapse while cardiovascular effort increases. Scores based on pace undercount this effort.
- **Environmental factors.** In heat, at altitude, or late in a long run, heart rate climbs while the external workload holds steady. This is known as cardiac drift, or decoupling [6]. A heart-rate score may read higher than the workload warrants.
- **Returning from injury.** Injuries often involve damage to a muscle, ligament or tendon. Measuring progress isn't about increasing cardiovascular effort but about carefully measuring biomechanical load, usually through a return-to-run or walk-run program.

3 Running power

The other established approach is running power, now offered on most sports watches and measured more directly by foot pods. A power model reduces a run's biomechanics to a single number in watts and derives training load from it in the tradition of power-based stress scores [2]. That load is measured against the runner's critical power: the highest power they can sustain without tiring, roughly what they could hold for 30 to 60 minutes. In simplified form, power scales with the runner's mass and speed, plus the work of climbing or descending:

$$\text{power} \approx \text{mass} \times \text{speed} \times (\text{cost of running} + g \times \text{grade})$$

Here the cost of running is the energy spent per kilogram for each meter on flat ground [10], g is gravity, and grade is the slope, which turns negative on a descent.

On a steep descent, estimated power drops: the runner produces fewer watts even as the quadriceps absorb heavy eccentric load, so they may receive an overall low score. On a hot day, power never sees the heart, so the environmental cost goes unrecorded.

The comparison cuts both ways. A foot pod records better raw biomechanics than any wrist sensor, and for steady-state running a power-based load has years of field use behind it. Because a foot pod measures each stride with its own motion sensors, its numbers depend on neither GPS nor heart rate. A runner gets consistent readings outdoors, on a level treadmill, or where the GPS signal is poor. In testing, the best-performing foot pod was the most repeatable of five running-power technologies both indoors and outdoors [11].

4 Introducing True Effort Score

True Effort Score (TES) captures each moment of a run as a vector of six signals from the watch: heart rate, pace, cadence, grade, vertical oscillation and altitude. TES keeps these signals together rather than reducing them to a single number at the start. TES ships as a machine learning model in Quiver, the open-source Swift framework.

A vector is a set of numbers treated as one unit. Each number represents one aspect of the moment, and together they position that moment in space. The distance between two vectors then shows their similarity: moments that look alike sit close together, moments that differ sit far apart.

A single number can make distinct efforts appear identical. A strong effort on flat ground and a rapid descent can share nearly the same pace, while their heart rate and vertical oscillation pull in opposite directions. As vectors, the two moments remain distinct:

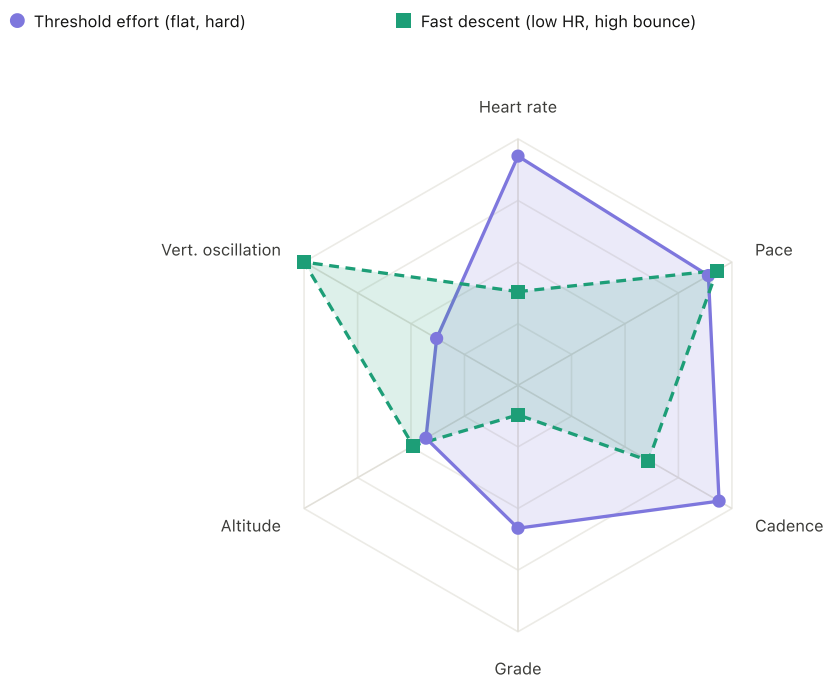


Figure 1. Two moments of a run as six-signal vectors: a threshold effort on flat ground and a fast descent. Values are illustrative and scaled to a common range.

TES also uses distance to classify effort. Its classifier, a nearest-neighbor model, labels each moment by finding the labeled examples closest to it. For more on vectors, see the Linear Algebra Primer in the Quiver documentation.

4.1 How it works

We ask two questions at each time interval. First, what should the heart be doing for this workload? A personal baseline, learned from the runner's own history, predicts the heart rate the workload should produce. The gap between this prediction and the actual reading is **the residual**: extra cardiovascular effort the workload doesn't explain (see The residual).

Second, what kind of effort does each moment look like? A classifier compares the movement pattern against known efforts and labels it **Easy, Tempo, Threshold, or Hard** (see Effort categories). The score adds up these labeled moments, weighted by intensity and duration, plus a few costs that only exist across a whole run (see The score).

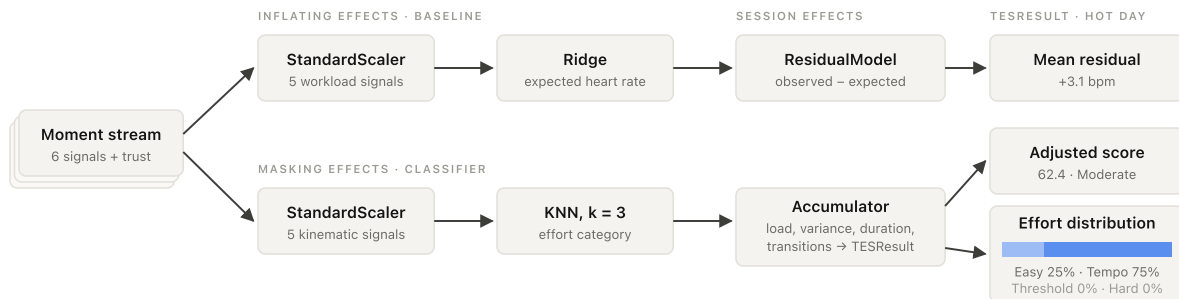


Figure 2. The two branches of TES: the baseline and the classifier.

Optical heart rate is not always reliable: a loose band, cold skin, or cadence lock, where the sensor tracks foot-strikes instead of heartbeats, can all distort it [8]. The app can mark each reading with how far it trusts it, and doubted heart rate counts for less.

4.2 What it solves

Traditional systems rely on a single signal, either heart rate or power estimated from speed, grade and mass, to interpret an entire run. TES works the other way around. It reads all six signals to label each moment first, then sums the labeled moments into the score.

This lets the scenarios described earlier function as intended. Descents and power-hikes count because the classifier evaluates the full movement pattern rather than one potentially misleading signal. Intervals count at full intensity because each moment is labeled individually, so a surge reads as a surge however easy the recovery around it. Finally, heat and cardiac drift are separated into a residual beside the score, because the baseline predicts the heart rate a given workload should produce.

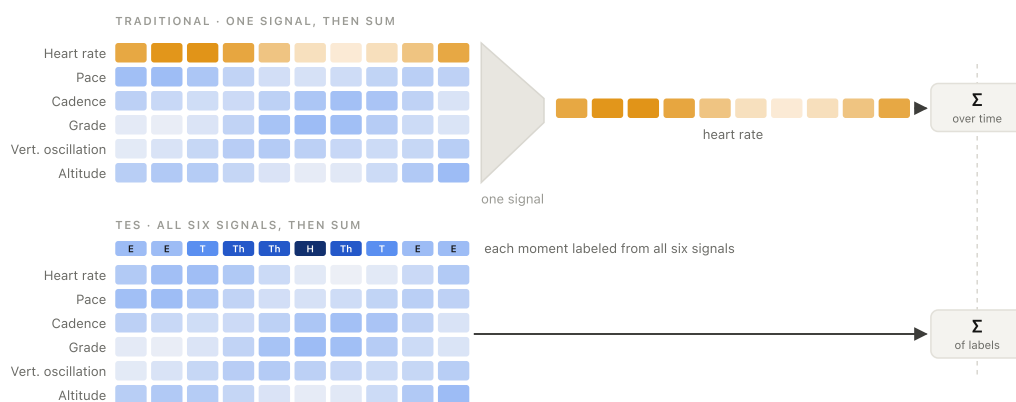


Figure 3. Traditional scores pick one signal — heart rate, or power derived from speed, grade and mass — then sum it over time. TES reads all six signals, labels each moment, then sums. Values are illustrative.

4.3 The baseline

The **baseline** learns the heart rate a runner's workload should produce, from the last four weeks of that runner's history. It is a ridge regression [5], a linear regression with a light penalty that keeps the learning stable when signals overlap, learned from **pace**, **cadence**, **grade**, **vertical oscillation** and **altitude**. Unlike other systems the calculation doesn't require knowledge of one's age, heart-rate zones or other demographic inputs. The learned baseline can be read directly. Learned from a synthetic 12-run history, this is expressed as:

$$\begin{aligned} \text{expected HR} = & 142.7 - 4.624 \text{ pace} + 2.871 \text{ cadence} + 4.853 \text{ grade} \\ & - 3.952 \text{ vertical oscillation} + 0.5405 \text{ altitude} \end{aligned}$$

The first number, 142.7, is this runner's average heart rate across their history: the heart rate expected when every signal sits at its usual value. Each **coefficient** then adds or subtracts beats per minute as its signal moves one standard deviation, a typical swing for this runner, above its

average. Because every signal is put on the same scale, their sizes can be compared directly. For this runner, grade counts about **nine times more** than altitude. Pace is negative because it is measured in minutes per kilometer, so a larger number means slower running and a lower heart rate.

4.4 The residual

By comparing this expected heart rate against measured heart rate, we can reveal the effects of other factors. A moment one standard deviation steeper than usual, with every other signal at its average, is expected to read $142.69 + 4.853 \approx 147.5$ bpm. If the watch reads 155 bpm at that moment, the gap of +7.5 bpm is the **residual**: heart rate the workload doesn't explain. TES reports it at every moment and as an average across the run.

The residual is reported beside the score rather than added to it, because the model cannot tell heat from caffeine or fatigue. A high score driven by real work and a high score driven by heat must remain distinguishable.

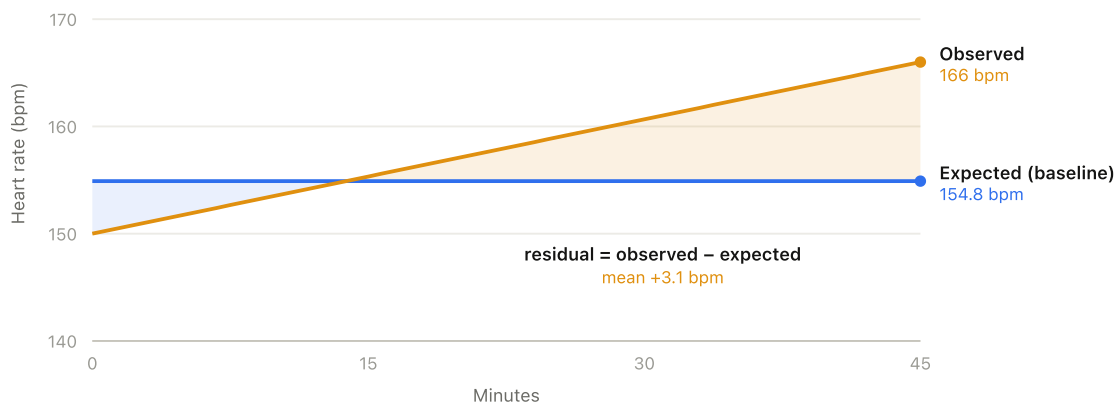


Figure 4. The hot-day scoring example: observed heart rate drifts above the baseline's expectation over 45 minutes. The shaded gap is the residual, mean +3.1 bpm.

4.5 Effort categories

The classifier is a nearest-neighbor model [3] that compares each moment with a set of known efforts and gives it the label of the closest ones. There are five categories:

Category	Meaning
Recovery	Walking, labeled by the walking gate rather than the classifier
Easy	Recovery and aerobic base
Tempo	Sustained sub-threshold, roughly marathon to half-marathon pace
Threshold	At lactate threshold
Hard	A heavy biomechanical load not reflected by heart rate or a sustained effort above threshold

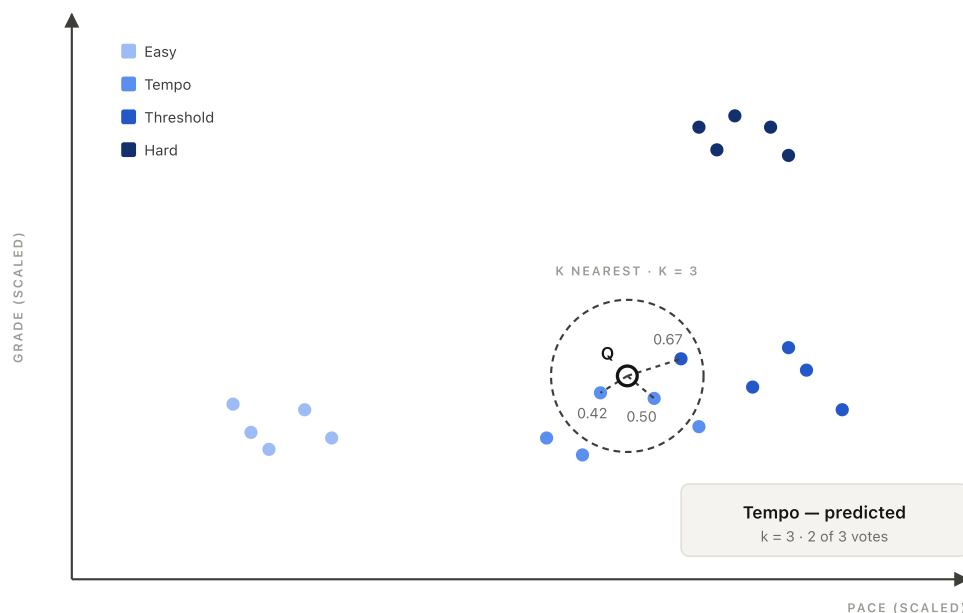


Figure 5. A new moment Q is labeled by majority vote of its three nearest known efforts: two Tempo and one Threshold, so the moment reads Tempo. Shown on pace and grade, two of the five scaled classifier signals; the model votes in the full five-signal space. Values are illustrative.

A moment on a steep descent at 131 bpm, 5.1 min/km and -5.8% grade reads **Easy** by heart rate alone. Its grade and vertical oscillation match the descent efforts in the known set, so TES labels it **Hard**.

The top category is intentionally called **Hard** rather than a cardiovascular zone name such as VO_2 max. A steep eccentric descent belongs in it because the legs are working at the top of their range while the heart is not. Naming the category after a cardiovascular quantity would contradict the reason the moment is there.

4.6 The score

Time in each category adds up to a score, and harder categories count more. The scale is anchored so that one hour held at threshold reads 100, following the convention of power-based training stress scores [2].

The category of **Threshold** is a real, repeatable effort a runner can hold for about an hour, rather than an abstract maximum. The score rises with both intensity and time and has no ceiling.

Some costs exist only across a run. **Intervals** cost more than steady running of the same average effort, and abrupt jumps between categories cost more than gradual changes, so both raise the score. Long runs get a small credit for time on feet past 45 minutes. With that credit, the same hour at threshold reads 102.9, the headline adjusted score. Each adjustment is also reported on its own, so a reader can see what moved the score away from the raw total.

The credit grows slowly on purpose. For runners covering 40 or more miles a week, the highly esteemed running coach **Jack Daniels** recommends **the lesser** of 150 minutes or 25 percent of weekly mileage [12]. At 150 minutes the credit reaches 12 percent. The larger cost of a substantial long run comes with adequate recovery, items not necessarily tracked with TES.

Values for easy running are low by design. An hour at **Easy** counts a third as much as an hour at **Threshold**, a steeper drop than power-based scores use: a two-hour easy run reads 73 in TES, where [2] would place it near 85. TES exists to price the work other scores miss, such as descents and surges, so its scale climbs most steeply at the hard end. Long runs still build the score through time, and the minutes spent in each category are reported beside it. These weights are design choices drawn from training-load practice [1, 2, 4, 7] that have not yet been tested against measured physiological cost.

4.7 What a score means

The adjusted score operates on an absolute scale divided into five ranges. Each boundary represents slightly more than one continuous hour in a specific effort category. Factoring in the time-on-feet credit, a steady hour yields a score of 34 at **Easy**, 69 at **Tempo**, 103 at **Threshold**, and 137 at **Hard**. As a rule of thumb, **every 35 points** represents one hour spent at the next highest intensity.

Range	Score	Equivalent Continuous Effort
Light	< 35	Up to one hour at Easy
Moderate	35–70	Up to one hour at Tempo
Substantial	70–105	Up to one hour at Threshold
Heavy	105–140	Up to one hour at Hard
Extreme	≥ 140	More than one hour at Hard

Table 1. Named ranges on the adjusted score.

Because surges and abrupt changes in category compound the score, mixed-intensity sessions often reach higher ranges in less time than steady efforts. These labels quantify the physiological cost of a workout; they are not recovery suggestions for the next day.

These boundaries are current design choices and remain provisional pending our validation study. To calibrate our expectations, we can look at two typical extremes. A recreational runner logging 45–70 minute easy runs will generally score between 25 and 40, straddling the Light and Moderate ranges. In contrast, a mountain runner sustaining high effort on climbs and descents will routinely land in the Heavy range, such as the 117.9 downhill example below.

4.8 Scoring examples

We scored four constructed runs through the model to demonstrate how these design choices behave in practice. Each example indicates the model state: *first run* (no personal history) or *established runner* (a learned baseline). These examples illustrate intended model behavior rather than physiological accuracy, which the planned validation study will test. The downhill and hot-day runs use a 5-second sampling interval, matching Apple's reported heart-rate frequency during workouts [8].

A downhill trail run (first run): 117.9, Heavy. A 55-minute run featuring four extended –6% descent blocks. While the runner's heart rate remains calm, movement data categorizes 58 percent of the run as Hard. A traditional heart-rate model would log this as an easy day; TES prices the eccentric load.

Hill repeats versus steady Tempo (first run): 76.4 versus 52.4. Both 47-minute sessions accumulate nearly identical raw load (52.8 versus 52.2). A basic metric of load multiplied by duration fails to distinguish them. TES successfully separates the two by pricing the surges.

A hot day versus a cool day (established runner): 62.4 versus 62.3. Both runs maintain the exact same steady pace, but cardiac drift on the hot day drives the heart rate from 150 to 166 bpm over 45 minutes. Because the work is the same, TES does not let the extra heart rate raise the effort category, so the score barely changes. Instead, the extra beats appear in the residual, the gap between actual and expected heart rate: +3.1 bpm on the hot day versus –2.8 bpm on the cool day.

Returning from injury: 10.9 to 15.6, Light. A runner recovering from a knee sprain follows a walk-run program. As their running time increases from 5 to 28 minutes over six weeks, their score gradually rises from 10.9 to 15.6. However, adding an 8-minute descent to that same 28-minute run jumps the score to 39.6 (**Moderate**), as grade and vertical oscillation label the downhill as **Hard** even though their heart rate stays calm.

Here, TES acts as a load advisor. While steady scores show a safe recovery, a sudden spike on the downhill warns of hidden stress that heart rate alone would miss.

5 Using TES

TES ships as part of Quiver, an open-source Swift package for statistics, linear algebra and machine learning, released under the Apache License 2.0. Quiver has no external dependencies, and the same code runs on iOS, macOS, watchOS, tvOS and visionOS, as well as on server-side Swift and Linux. Add it with Swift Package Manager:

```
dependencies: [  
    .package(url: "https://github.com/waynewbishop/quiver", from: "1.6.0")  
]
```

`TrueEffortScore` is constructed empty for a new runner, or decoded to resume a saved one. Samples are recorded live, and `finalize()` closes the run.

```
import Quiver  
  
var tes = TrueEffortScore()           // first run: no personal history yet  
  
// For each sample, live or read back from HealthKit after the run:  
tes.record(heartRate: 152, pace: 5.4, cadence: 174, grade: 1.2,  
           verticalOscillation: 7.9, altitude: 100,  
           at: sampleDate, hrTrust: 1.0)  
// bpm, min/km, steps/min, percent, cm, meters; hrTrust is the  
// app's confidence in the heart-rate reading, 0 to 1  
  
tes.currentScore           // live score, before session adjustments  
tes.currentEffort          // .easy, .tempo, .threshold or .hard  
                           // (.recovery only for detected walking)  
tes.currentResidual        // observed - expected HR, once a baseline exists  
  
if let result = tes.finalize() {      // nil if the run is too short to keep  
    result.adjusted           // the headline score  
    result.raw               // before session adjustments  
    result.meanResidual      // average residual across the run, in bpm  
    result.expectedHeartRate // average expected HR, nil on a first run  
    result.effortDistribution // share of time in each category  
}  
  
if let baseline = tes.baseline {      // nil until a baseline has been learned  
    baseline.equation()  
    // "expected HR = 142.69 - 4.62·pace + 2.87·cadence + 4.85·grade  
    //                   - 3.95·verticalOscillation + 0.54·altitude"  
}
```

6 Roadmap

Upcoming development for TES includes:

Area	Planned work
Walking and hiking	Refining the scoring model for recorded walks and hikes
Treadmill incline	Adding manual incline entry to properly score indoor hill sessions
Personalized heart rate	Adapting scores to each runner's own heart-rate range
Hot days on the first run	Scoring hot days accurately from a runner's very first run
Unfamiliar elevation	Adjusting scores for runs at altitudes outside the runner's history

Table 2. Planned development for TES.

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